

The rotation-reptation transition under broken rotational symmetry

A. Feltrup¹, K. Huang¹, C.A. Krülle², and I. Rehberg¹

¹ Experimentalphysik V, Universität Bayreuth, 95440 Bayreuth, Germany

² Maschinenbau und Mechatronik, Hochschule Karlsruhe – Technik und Wirtschaft, 76133 Karlsruhe, Germany

Abstract. In a swirled circular container, granular particles can change their sense of rotation when the packing density is increased, exhibiting a transition from rotational to reptational motion. In addition, here we report a ‘snake mode’ that arises at a lower packing density, where particles form a chain like cluster that rotates with the same frequency as the container. We investigate experimentally transitions between these three modes under the influence of geometrical distortions which break the rotational symmetry of the container. The driving mechanism for the rotational motion of the clusters is also discussed.

Due to its ubiquity in nature, industry and our daily lives, granular matter has attracted interest from both engineering and scientific communities over centuries [1–3]. Because of the energy dissipation through internal friction and inelastic collisions between particles, agitated granular matter is generally speaking in a non-equilibrium state. Therefore it acts also as a model system to study systems far from equilibrium. In the past decades, many interesting phenomena of granular matter under the driving of vertical or horizontal vibrations have been discovered, for instance pattern formation [4,5], localized excitation [6,7], and size or density segregation [8–10]. For granular matter, we are now able to determine the movements of particles with the aid of high speed photography, which enables us to explore this non-equilibrium system at both ‘microscopic’ and ‘macroscopic’ levels [11,12]. In addition, computer simulations could be employed to explain various phenomena including pattern formation, cluster formation, and inelastic collapse [4,13–15].

Here we focus on a quasi two-dimensional system driven by horizontal swirling. A motion not only frequently used to stir up the bouquet of a glass of wine, but also to agitate the particles in vibrational mills [16,17]. Moreover this motion proved to be able to enhance the efficiency of pneumatic conveying [18], which is widely applied in many chemical engineering and industrial processes. Such a quasi two-dimensional system is conceptionally simple since gravitation plays no role. As a technical advantage, here one is able to trace all particles with the help of image processing procedures.

Former studies on such a system reveal that the sense of rotation of the granular materials changes with the packing density [19]. At lower packing densities, the cluster rotates with the same direction as the swirling container (rotation mode). At higher packing densities, the rotation changes its direction and the diffusivity of particles within the cluster is dramatically reduced (reptation mode). This is a robust phenomenon and frequently observed in various industrial processes, for example in the vibrational mills for grinding feed materials [16,17]. Further studies indicate that this transition occurs also in a one dimensional swirling annulus [20]. In a follow-up study [21], the particles are found to be embedded into a solid-like cluster in the reptation mode. This is discovered by recording the trajectory of a tracing particle and

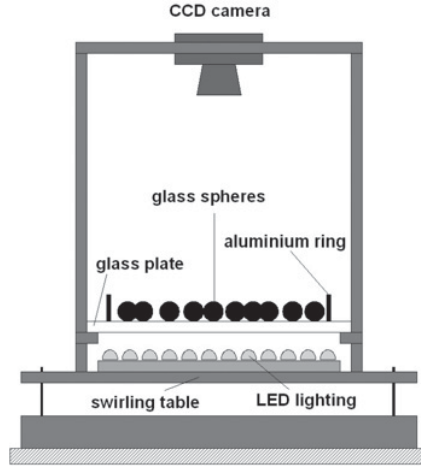


Fig. 1. Sketch of the experimental setup.

compare the density and temperature in different modes. Later on, similar image processing techniques are employed to study horizontal particle segregation, the Brazil-Nut effect and its inverse [22–24].

To understand the transitions between different modes of the cluster rotation, it is necessary to know the driving mechanism for the clusters. For the swirling table, one might speculate that the rotational symmetry of the driving wall is essential for the rotational motion of the clusters. Based on this speculation, we raise the question how stable those modes are under a broken rotational symmetry. In order to answer this question, we introduce one or two obstacle particles into the container in a systematic way and measure the rotational velocity of clusters.

A device similar to the one used in Ref. [19] is used for the experiments, as sketched in Fig. 1. The container consists of an aluminium ring with an inner diameter of 10.5 cm mounted on a glass plate. It is vibrated horizontally, performing a so called *swirling* motion. It means that every point of the container is moving along a circular path described by $x = A \cos(2\pi f_d t)$ and $y = A \sin(2\pi f_d t)$, where A is the amplitude and f_d the driving frequency. In our experiment the amplitude of the motion is set to 1.59 cm and the driving frequency to 1.67 Hz in a counterclockwise direction (viewed from above). The dish is filled with glass beads with a diameter $D = 10.0 \pm 0.02$ mm and a density of 2.5 g/cm^3 . The maximum number of spheres which can fit into a monolayer is 88. To disturb the rotational symmetry obstacles made out of aluminium spheres with a diameter of 10 mm are fixed to the glass plate. The granular sample is illuminated from below by a LED array. A charge coupled device (CCD) camera is mounted on top of the swirling table to take pictures in the comoving frame. The images captured by the camera are then subjected to an image processing procedure to locate and trace all the particles in the container. The particle tracking is based on the Hough transformation adapted to detect circular objects [25]. After this procedure, we are able to obtain both the position and the velocities of all the particles at all time for different measurements.

Without any fixed obstacles, we observe three different kinds of motion of clusters depending on the number of particles inside the dish. Visualization of cluster motions in different modes are shown as insets in Fig. 2(a). The trajectories of randomly selected beads are superimposed on the snapshots taken by the camera. At low particle numbers all beads travel to the border after a short time, forming a snake-like cluster which travels along the container wall with the same frequency and direction as the driving. The formation of this so called snake mode, first observed by M. Raithel, depends on the size of the dish and couldn't be observed in larger dishes [26]. The dependency of dish radius and driving parameters on the formation of this mode is still under investigation. When the particle number is higher than a certain value, the beads start to pile up in the radial direction, entering a so-called rotation mode. The cluster still rotates counterclockwise but with a lower frequency than the driving. As the particle trajectories indicate, the motion of particles within the cluster is more randomized. Further

increase of the particle number slows down this rotation, until at a critical particle number N_c the rotation stops. Beyond this particle number, the cluster starts to rotate in the opposite direction, performing a motion called reptation. From the hypocycloidal trajectories, one could assume that the granular matter is in a solid-like state [21]. A description of the second and third motion was already presented in Ref. [19].

To get a quantitative expression of this behavior, we look at the cluster rotation frequency f around its center of mass, which is calculated as:

$$f = \frac{1}{2\pi N} \sum_{i=1}^N \left\langle \frac{\mathbf{r}_i^* \times \mathbf{v}_i^*}{|\mathbf{r}_i^*|^2} \right\rangle_t \quad (1)$$

with $\mathbf{r}_i^*$ and $\mathbf{v}_i^*$ the positions and velocities of the particles relative to the center of mass and $\langle \rangle_t$ the time average for each particle.

$$\mathbf{r}_i^* = \mathbf{r}_i - \frac{1}{N} \sum_{j=1}^N \mathbf{r}_j \quad (2)$$

$$\mathbf{v}_i^* = \mathbf{v}_i - \frac{1}{N} \sum_{j=1}^N \mathbf{v}_j. \quad (3)$$

This rotation frequency is then rescaled by the driving frequency f_d to get the normalized cluster rotation frequency f_c

$$f_c = \frac{f}{f_d}. \quad (4)$$

Figure 2(a) shows the normalized cluster rotation frequency as function of the particle number without any fixed obstacles. The transition from snake to rotation mode happens at $N = 17$. Further increase of the particle number leads to a drop of the rotation frequency to around 0.4 of the driving and in this mode particles no longer move in a collective manner. For the particle number range between 23 and 46, a plateau regime is observed, where the cluster rotation is insensitive to the number of particles in the container. When the number of particles is increased further, the cluster slows down until at $N_c = 66$ the rotation changes its direction and the cluster enters a reptation mode.

The effect of breaking the rotational symmetry by inserting fixed obstacles into the system can be seen in Fig. 2(b). If the obstacle is fixed at the center of the dish, the cluster rotation is almost undisturbed. One interesting feature is that in the rotation or reptation mode, the cluster rotation frequency is slightly higher than without any obstacles and the rotation-reptation transition increases to $N_c = 69$. This reveals that the system is more favorable to the rotational mode in this case. If the obstacle is placed at a distance of $3.65D$ from the center of the container, leaving a gap of 1.1 particle diameters between the obstacle and the aluminium ring, the rotation-reptation transition reduces to $N_c = 63$ and the snake mode ends at $N = 15$. If we move the obstacle further to the border, a slight shift of the rotation-reptation transition to lower particle numbers can be observed as the critical value of $N_c = 61$ indicates. The snake mode is suppressed completely, which can be understood as the traveling path of the snake is blocked by the obstacle. By inserting a second obstacle at the opposite side of the container wall, not only the snake mode but also the rotation mode is nearly suppressed too, the critical particle number for the rotation-reptation transition is found at $N_c = 18$ for this configuration. This indicates that the reptation mode becomes more favorable.

To understand the different rotation frequencies below the critical particle number N_c , we look at the driving mechanism of the system. One driving source comes from collisions of the beads with the border of the dish, the other comes from the bottom of the container. In Fig. 2(c) we plot the ratio between the number of particles at the border and the overall number of particles in the container as function of the particle number for all configurations shown in Fig. 2(b). Without obstacle or with one obstacle at container center, the ratios show a constant

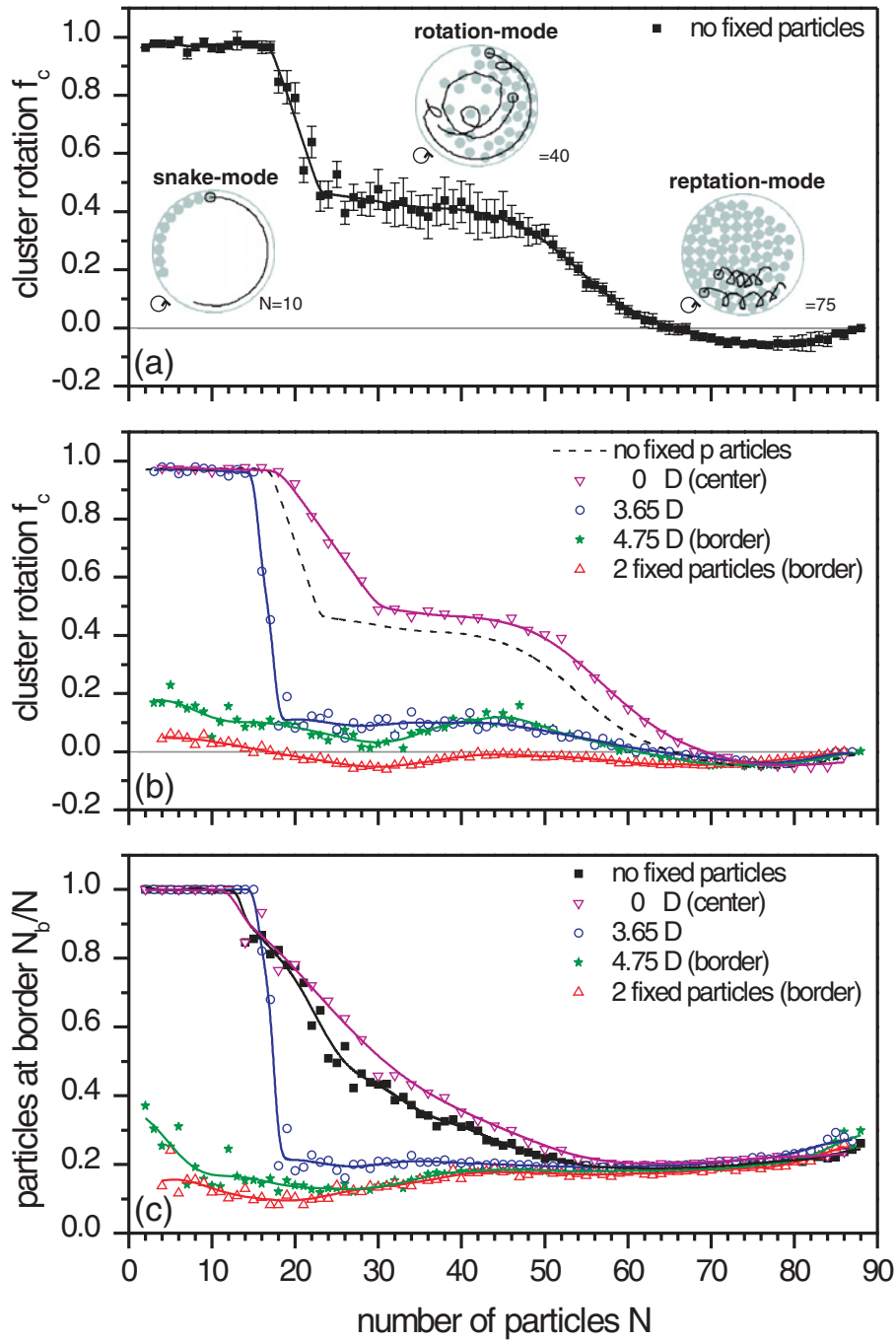


Fig. 2. (a) Rotation frequency of the cluster as a function of the number of particles N without fixed obstacles. Snapshots of images captured at different modes are shown as insets. Solid lines correspond to particle trajectories shown as examples. Particles surrounded by circles indicate the ends of the tracks. (b) Cluster rotation frequency as function of N with fixed obstacles. Different symbols correspond to different gap width between the obstacle and the container boundary in units of particle diameter D . With two fixed particles both are fixed at the boundary at opposing sides of the dish. The dashed line is a replot of (a) for comparison. (c) Ratio between particles at the border and the total number of particles as a function of N for the same configurations as above. The lines in all the plots are guide to the eyes.

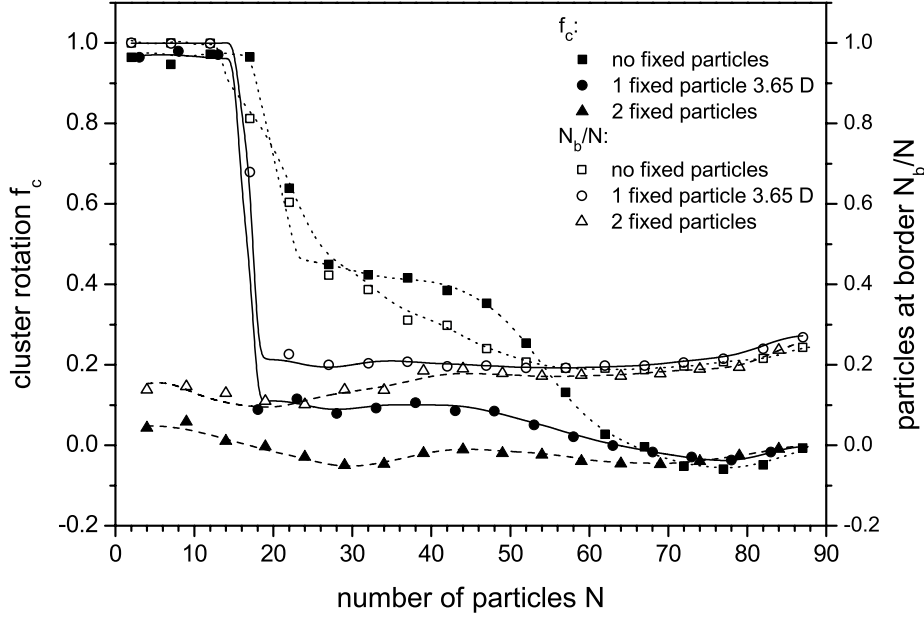


Fig. 3. Comparison between the cluster rotation and the fraction of particles at the border for three different configurations. The lines are guide to the eyes.

value of 1 for low particle numbers and thereafter a slow decay with the increase of particle number until they reach a fairly constant value of 0.2 at particle numbers of about $N = 60$. For the configuration with a gap that allows only one particle to pass through (corresponding distance from the center of the container $3.65 D$), a sharp drop of the ratio is observed at the transition to the rotational mode happens. For the configurations with obstacles at the border we always have a fairly constant particle ratio of about $20 \pm 10\%$ at the border. For all configurations shown here, similarities between the rotation frequency and the fraction of particles at the border suggest that the driving of rotation mainly stems from the interaction from particles at the border with the container wall.

One might notice that for the configurations without any obstacles or with one obstacle at the center, the drop in the particle fraction at the border begins while the system is still in the snake mode according to the rotation frequency. This can be understood by the fact that at $N \geq 13$ particles can be found in the second row adjacent to the snake travelling with the same frequency as the snake itself. This results in a drop in the particle fraction while leaving the rotation frequency unchanged. In the case that the gap between obstacle and the container wall only allows one particle to pass through, this effect is suppressed so that the drop happens at the same particle number.

To have a direct comparison between the normalized cluster rotation frequency and the fraction of particles at the border, we replot both of them for 3 different configurations in Fig. 3. The correlation between the two curves for each configuration indicates that the main driving force for the rotation of the cluster comes from the border of the container, while the bottom of the container plays a minor role.

To summarize, we study the rotation of swirling quasi two-dimensional granular clusters under a distorted geometry. Depending on the packing density, granular particles might rotate in a snake mode, rotation mode or reptation mode. The rotation frequency of clusters in the snake mode and the rotation mode depends sensitively on the distance between the obstacle and the container wall. When the obstacle is close to the center of the container, it indeed speeds up the rotating motion. This observation might have potential applications in industry to improve the efficiency of transport or mixing processes. In contrast, if the obstacle is fixed closer to the container border, the reptation mode is favored. Once the gap between the obstacle and container border is smaller than a particle diameter, the ‘snake mode’ vanishes and the rotation

frequency is dramatically reduced. The dependance of the rotation frequency of clusters on the particle number at the container border reveals that the rotation motion of clusters is mainly driven by the container wall.

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